ANAND PAGNIS

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EDUCATION

Worcester Polytechnic Institute (WPI) Worcester, MA | May 2026
Bachelor of Science in Computer Science, Minor in Fintech

TECHNICAL SKILLS

Frontend: React, TypeScript, JavaScript, HTML, CSS, Next.js, Reusable Component Libraries, Zustand, Three.js, GLSL
Backend and Data: Python, FastAPI, REST APIs, Node.js, PostgreSQL, MySQL, Supabase, SQL, C#, C/C++, Bash
Practices: AI-Assisted Development, Test-Driven Development, Git, GitHub Actions CI, Docker, Agile and Scrum, Jira, Code Review

EXPERIENCE

Buffeat, *Software Engineer*, *Frontend Developer* Remote | Sep 2025 - Nov 2025

- Built reusable component libraries in React and TypeScript for the core platform that consumed type-safe RESTful APIs, increasing feature development throughput by 20 percent.
- Developed, tested, and integrated component-based interfaces following Git workflows and object-oriented design, partnering with backend engineers to define FastAPI endpoints and delivering sprint commitments through code reviews.

SquatchScout, *Co-Founder* Remote | Apr 2026 - Jul 2026

- Built a full-stack, two-sided local-services marketplace in Next.js, React, TypeScript, and Supabase, with three-tier role-based access control enforced across middleware, server-side guards, and database policies.
- Delivered real-time scheduling, live chat, and a payment abstraction, with a GitHub Actions CI pipeline validating access control, checkout, and authentication on every pull request.
- Onboarded beta customers and gathered feedback on interface and functionality to refine the product.

Second Brain, *Co-Founder* Remote | Dec 2025 - Apr 2026

- Co-founded a local-first knowledge-retrieval system with a browser extension that captures articles and a 10-stage hybrid retrieval pipeline (BM25, pgvector ANN, cross-encoder reranking) running under 500 ms CPU latency.
- Designed a 9-table PostgreSQL schema, deployed on Railway with Docker, Supabase JWT authentication, and LRU caching.

Fundesteam, *Project Lead and System Administrator* Panama City, Panama | Aug 2024 - Oct 2024

- Led development of a low-power, scalable offline educational server for rural Panama with the Ministry of Education, improving video delivery performance by 85 percent through iterative profiling and optimization with Python, Docker, and Linux.

PROJECTS

AudioVis *(solo, in progress)* 2026 - Present

- Building a browser-based audio-visual performance engine that analyzes a song's structure in real time and drives GPU-raymarched shaders and particle systems, using React, TypeScript, Three.js, GLSL, Web Audio API, and Zustand.

Hospital Navigation Web Application, *Project Manager and Frontend Developer* Jan 2024 - Apr 2024

- Led a 9 member Agile team at WPI and Brigham and Women's Hospital building a hospital navigation and service app with React and TypeScript, using Jira and code reviews, deployed on AWS EC2 with Docker, Prisma ORM, and PostgreSQL.

ADArC: Automated Design Tool for Arduino Circuits (adarc.app) Aug 2025 - May 2026

- Expanded an Arduino circuit simulator by 25 plus components using C#, .NET Core, Blazor, TypeScript, AVR8js, and GraphSynth, with reusable, testable module design and real-time SVG visualization.

Algorithmic Trading Engine 2024

- Built a Python pairs trading engine analyzing S&P 500 equity price time series with a Streamlit and Plotly dashboard; won a university-wide, 3 month trading competition by outperforming the S&P 500 by 2.5 percent.

SEC 10-K Research Terminal May 2026

- Built a full-stack data platform with a Python and FastAPI backend and a React and TypeScript frontend, designing type-safe REST APIs across 15 plus data models, used by the Investing Association fund to develop reports.
- Used agentic development for boilerplate, debugging, and scaffolding, with test-driven and behavior-driven development.

LEADERSHIP

Investing Association, *President* Worcester, MA | Jan 2025 - Mar 2026

- Led a 250 plus member club on a 10,000 dollar budget and launched a six-figure student-led investment fund.

Alpha Chi Rho, *Social Chair* Worcester, MA | Aug 2024 - May 2025

- Led a 10 member committee to plan professional and social events for 100 plus students.